CHAPTER 12

NATIONAL INCOME

Learning Objectives

The students will learn:

- *The circular flow of income in two-sector, three-sector and four-sector models.*
- *The basic terms associated with the calculation of national income.*
- *The calculation of national income by value-added, income and expenditure methods.*
- *The difference between GDP and welfare.*

Course of Action
Maximum Time = 4 Hrs
Maximum Questions = 93

Topic-1
Circular Flow of Income **Page No. 238**

Topic-2
Basic Concepts of National Income Accounting **Page No. 243**

Topic-3
Measurement of National Income **Page No. 249**

Topic-1 Circular Flow of Income

Concepts covered: Principles of Circular Flow of Income, Real Flow, Money Flow, Conditions of Equilibrium

Scan to know more about this topic

National Income - Circular Flow of Income

Revision Notes

- **Circular Flow of Income :** A continuous flow of production, income and expenditure in an economy is known as circular flow of income. It is circular because it has neither any beginning nor an end.
- **Principles of Circular Flow of Income :** (1) The seller receives the same amount as spent by the buyer, and (2) The direction of flow of goods and services is opposite to the direction of flow of money spent to acquire those goods and services.
- **Real Flow :** It shows the flow of goods and services among the various sectors of economy.
- **Money Flow :** It shows the flow of money among various sectors of economy.
- **Conditions of Equilibrium :**
  - **Two Sector Model**

    $S = I$

    Or

    $C + S = C + I$
  - **Three Sector Model :**

    $C + S + T = C + I + G$

  (iii) Four Sector Model :

    $Y = C + I + G + (X - M)$
- **Four Sectors of the Economy :**
  - Household sector
  - Firms or producer sector
  - Government sector
  - External sector (Rest of the world)
- **Leakages of Income :** That part of income which is not returned back to circular flow is called leakage of income. For example, Savings, Tax and Imports, etc.
- **Injections of Income :** Some additions made in the economy from sources like investment, government expenditure and exports.

National Income

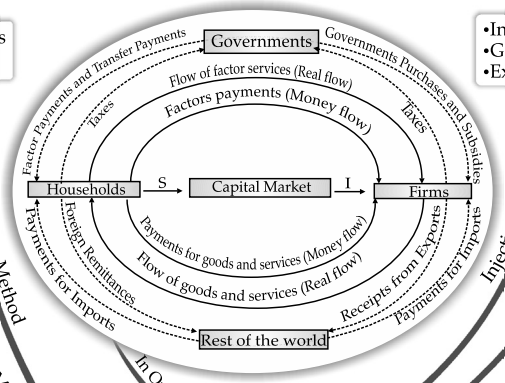
Calculating National Income

- Factor Income Method
  - NDP_{fc} = Compensation of Employees + Operating Surplus + Mixed Income $NNP_{fc} = NDP_{fc}$ + NFIA
- Output or Value Added Method
  - GDP_{mp} = value of final output - intermediate consumption $NNP_{fC} = GDP_{mp}$ - Depreciation- Net Indirect Tax
- Expenditure Method
  - Private final Consumption Expenditure + Government Final Consumption Expenditure + Gross Domestic Capital Formation(GDCP)+ Net exports(x-m)
- At market price
  - GDP_{mp}
  - $GDP_{fc} = GDP_{mp}$ + NIT
  - $NDP_{mp} = GDP_{mp}$ – Depreciation
  - $NNP_{mp} = NDP_{mp}$ + NFIA
- At factor cost
  - NDP_{fc}
  - $GDP_{fc} = NDP_{fc}$ + Depreciation
  - $GNP_{fc} = GDP_{fc}$ + NFIA
  - NNP_{FC} = NI = GNP_{fc} – Depreciation

National Income Aggregates

- Private Income
  - NI–Income from property and entrepreneurship accruing to government Administrative department– Savings of Non Departmental Enterprises of Government + Interest on National Debt + Net Current Transfers (CT) from Government – Net CT from Rest of the World
- Disposable Income Aggregates
  - Personal Income
    - Private Income – Undistributed Profits – Corporate Profit–Retained Earnings of Foreign Companies
  - Personal Disposable Income
    - Personal Income- Personal Taxes- Misc. receipts of Government Administrative Departments
- Other Concepts
  - Real GDP
    - GDP at Constant prices
  - Nominal GDP
    - GDP at Current prices
  - GDP Deflator
    - $\frac{\text{GNP at Current Price}}{\text{GNP at Constant Price}} \times 100$

Circular Flow of Income



- In Open Economics
- Injections
  - •Investment •Government Expenditure •Exports
- Leakages
  - •Saving •Taxes •Imports

Basic Concepts

- Domestic territory
  - is geographical area administered by Government of country
- Normal Residents
  - a person or institution whose economic interest lies in that country
- Transfer Payments
  - unilateral payments without any value addition in economy. e.g., Gifts, donations
- Consumption Goods
  - purchased for satisfaction of want, e.g., TV, Pizza
- Capital Goods
  - goods used in production of other, e.g., plant, machinery, etc
- Intermediate Goods
  - goods which are used by producer as inputs into a further stage of production, also known as 'non factor inputs'
- Final Goods
  - purchased for the purpose of final consumption e.g; car, machine, etc
- Stocks and Flows
  - flows are defined over a period of time whereas stocks are defined at a point of time
- Investment
  - Net Investment
    - Gross investment– Depreciation
  - Gross Investment
    - The part of the final output that comprises of capital goods constitutes gross investment
- Depreciation
  - Loss in the value of fixed assets/capital goods or consumption of fixed capital

GDP & Economic Welfare

- As an Indicator of Economic Welfare
  - GDP is regarded as index of welfare as:-
    - Economic welfare increases with the expansion in GDP
    - As the GDP expands, per capita income of the persons also increases
    - Expansion of GDP enhances the infrastructure of the country
- Limitations of GDP as Indicator of Economic Welfare
  - Expenditure on Regrettable Necessities
    - •Expenditure forced on consumer. •e.g., Expenditure on medical facilities
  - Monetisation of Goods and Services
    - •Non-inclusion of non-monetised goods •e.g., Services of house values, mother, etc
  - Exclusion of some Items from GDP
    - e.g., Leisure, non-market transactions, Quality of life etc.
  - Manner of Increase in National Income
    - e.g., If increase in GDP is result of increased employment of child labour
  - Consumption of national Income
    - e.g., Per capita income has increased due to increase in total output by production of more of non-civilian and capital goods
  - Distribution of National Income
    - Inequalities in distribution of income and wealth

Trace the Mind Map

First Level · Second Level · Third Level

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Give the meaning of nominal flow. [OEB]

Ans. Nominal flow can be referred to as that flow in which factor payments flow from firms to households and consumption expenditure is made by households on goods and services produced by firms.

Q.2. Mention any two sectors of the economy. [OEB]

Q.3. State any one difference between real flow and money flow. [OEB]

Ans. Real flow involves flow of goods and services and factor services whereas money flow involves flow of factor payments and consumption expenditure.

Q.4. Name the sectors in the circular flow of Income in a three-sector model. [OEB]

Ans. Household, Production and Government.

Q.5. Name the sectors in the circular flow of Income in a four-sector model. [OEB]

Ans. Household, Production, Government and Foreign.

Q.6. State whether the following is True or False. Give one reason for your answer.

(a) Change in inventory is a stock concept. [ISC Specimen Paper, 2023-24]

Ans. False

Reason: Change in inventory is a flow concept as it involves the inventory to flow.

(b) "Savings and Taxes are injections of circular flow of income."

Ans. False

Reason: This is because savings and taxes are leakages to the circular flow of income.

(c) "Investment and government expenditure are leakages to the circular flow of income."

Ans. False

Reason: Investment and government expenditure are injections to the circular flow of income.

(d) "In a two-sector model of circular flow of income, there are two sectors government expenditure and rest of the world."

Ans. False

Reason: In a two-sector model of circular flow of income, there are two sectors consumption expenditure and investment expenditure.

(e) "In real flow the factor payments like rent, wages, interest and profit flow from firms to households."

Ans. False

Reason: This is because in money flow, factor payments such as rent, wages, interest and profit flow from firms to households for the factor services in the form of land, labour, capital and entrepreneur made by the households.

(f) "In money flow, factor services flow from firms to households."

Ans. False

Reason: In money flow, factor payments in return of factor services provided by the households **is given to the households.**

(g) "In a national income equilibrium, there are two conditions, saving = consumption and consumption + saving = consumption + government expenditure."

Ans. False

Reason: In national income equilibrium, saving is equal to investment and aggregate demand (consumption + Investment) = Aggregate supply (consumption + saving).

2 Marks Questions

Q.1. With the help of a diagram, show the circular flow of income in a two-sector model with savings and investment. [ISC 2023]

OR

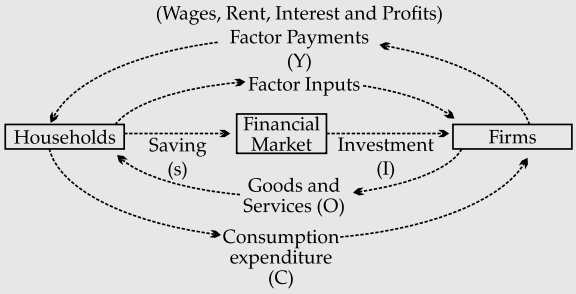
Draw a diagram to show the circular flow of income in a two sector model withleakage and injection. [ISC 2017]

Concept Applied	Circular flow of income

Ans. The circular flow of income is a way of representing the flows of money between the two **main groups in society - producers (firms) and consumers (households)**. These flows are part of the fundamental process of satisfying human wants.

As we have already seen, a free market economy consists of two components, or sectors, as they are called. These are firms and households. People in households work for firms (selling their factor services) and receive wages in exchange. On the scale of the whole economy, this is known as national income - the total amount of income earned over a given time period. This money is spent on food, clothing, transport, entertainment etc, and so it returns to the firms.

This is the circular flow.



This question is for practice and its solution is available at the end of the chapter

Leakage and injection in two sector model:

Households sell their factor services to firms (in the factor markets) and in exchange receive wages (the left hand side of the flow). In the meantime, households spend this income on goods and services (in the goods market) and in exchange receive the goods and services themselves (the right hand side of the flow). Domestic consumption spent, however Some is saved. Savings are coded as 'S'.

'S' is called leakage from the circular flow. The effect of these leakages can be seen in the figure.

Leakage:

A leakage reduces the flow of income. On the other hand, some firms borrow money and invest it in their firms in the form of capital goods. This is 'I' for investment called injections as the money returns into the circular flows.

[ISC Marking Scheme, 2017]

Examiner's Comment

- In this part, several candidates did not depict capital market, savings and investment in the circular flow of income. Leakages and injections were neither shown in the diagram nor explained by a large number of candidates.

Answering Tips

- Students need to practice drawing circular flow of income with correct directions.
- Learn the relevance of Leakages and Injections for savings and investment respectively.

Q.2. Give difference between stock and flow concept with examples? [OEB]

Ans.

	Stock	Flow
1.	It can be referred to as that variable which is calculated at a particular point of time.	It can be referred to as that variable which is calculated for a period of time.
2.	For example- Money supply, national wealth.	For example- National income, government expenditure.

Q.3. Mention the difference between leakages and injections in a circular flow of income model? [OEB]

Ans.

	Leakages	Injections
1.	It can be referred to as the money withdrawn from the circular flow of income model.	It can be referred to as the money injected in the circular flow of income model.
2.	For example- Saving, taxes, Imports are the leakages.	For example - Investment, government expenditure and exports are the injections.

Q.4. Classify the following as stock or flow: [OEB]

(i) Change in supply. **(ii) Inventory.**
(iii) Capital. **(iv) Depreciation.**

Ans. **(i)** Flow **(ii)** Stock
(iii) Stock **(iv)** Flow

Q.5. Classify the following as leakages and injections: [OEB]

(i) Savings **(ii) Investment.**
(iii) Exports **(iv) Imports**

Ans. **(i)** Leakages **(ii)** Injections
(iii) Injections **(iv)** Leakages

3 Marks Questions

Q.1. With the help of relevant examples, explain the meaning of leakages and injections in the circular flow of income. A I [ISC 2020]

Ans. Leakages mean to withdraw money from the circular flow of an economy. Leakage from the circular flow of income of an economy happens when the firms and households save a part of their incomes. Therefore, leakage or withdrawal is that part of the income of an economy that does not pass through the circular flow of income, resulting in the unavailability of that money for spending on the goods and services produced recently. Thus, it can be said that leakages reduce the flow of income in an economy.

Examples: Saving, tax, imports.

Injections mean the addition or introduction of income to the circular flow of an economy. Injections into the circular flow of income are a result of money borrowed by households and firms from different external sources, like financial institutions. However, this additional income does not result in an immediate expenditure. Therefore, injections increase the flow of income in an economy.

Examples: Investment, public expenditure, exports.

Examiner's Comment

- Many candidates were not able to state the meaning of leakages and injections properly. Some candidates drew the diagram of circular flow of income for two- sector, three-sector and four sector models, but did not indicate saving, tax and imports as leakages and investment, public expenditure and exports as injections.

Answering Tip

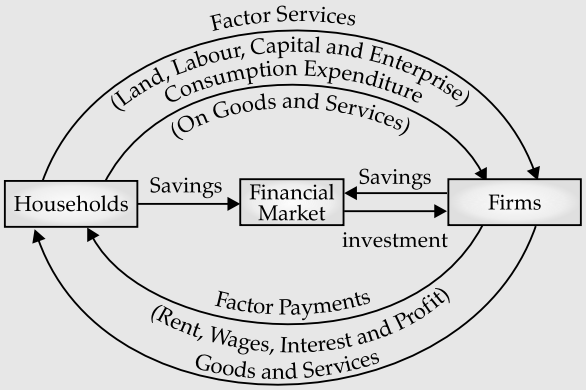
- Practice the diagrams of two and three sector models of circular flow separately, mentioning the leakages and injections in each. Ensure that it is well understood that why a particular term is a leakage or injection and how each one of them affects the circular flow of income.

Q.2. Draw a well-labelled diagram to show a circular flow of income in a two–sector model. What happens to the flow of income when savings equals investment? [ISC 2019]

OR

With the help of a diagram, show the circular flow of income in a two sector model with Savings and Investment. [ISC 2015]

Ans.



Equilibrium in the two sector model of circular flow of income will take place when S=L, saving constitutes a leakage, and investment is an injection in the circular flow of income.

Contraction of the level of income resulting from savings is just equal to the expansion of level of income resulting from the act of investment.

[ISC Marking Scheme, 2019]

Examiner's Comment

- Several candidates did not explain the saving-leakages and investment-injection for circular flow of income. The answer to 'what happens to the flow of income when saving equals investment' was missing in many answer scripts. Many candidates did not show the directions of the flow which made the diagram meaningless. Some candidates did not show the difference between real flow and money flow.

Answering Tips

- Clearly understand the meaning of leakages, injections, real flow and money flow while teaching the circular flow of income in the two-sector model.
- Analyse the stage of S=I and other relations.

Q.3. Draw a diagram for circular flow in a three - sector model. [OEB]

Q.4. State the assumptions on which the circular flow in a two sector model is based upon. [OEB]

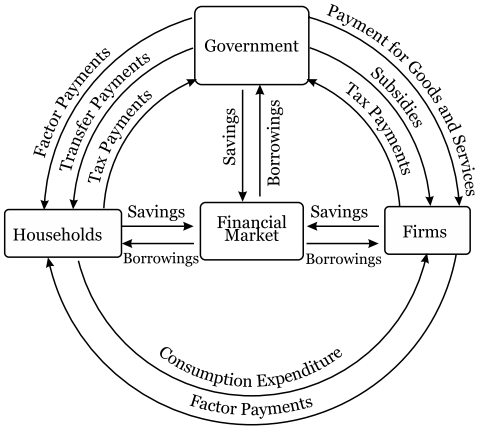
Ans. The assumptions are:

(i) There are only two sectors in the economy; that is, household and firms.
(ii) Households supply factor services to firms.
(iii) Firms hire factor services from Households.
(iv) Households spend their entire income on consumption.
(v) Firms sell all that is produced to the households.
(vi) There is no government or foreign trade.

4 Marks Questions

Q.1. Draw a well labelled diagram to explain briefly, the circular flow of income in a three-sector model. [OEB]

Ans.



Explanation: Three sector model includes household sector, producing sector and government sector. It will study a circular flow income in these sectors excluding rest of the world i.e., closed economy income. Here flows from household sector and producing sector to government sector are in the form of taxes. The income received from the government sector flows to producing and household sector in the form of payments for government purchases of goods and services as well as payment of subsides and transfer payments. Every payment has a receipt in response of it by which aggregate expenditure of an economy becomes identical to aggregate income and makes this circular flow unending.

Q.2. State giving reason whether the following are stock or flow: [OEB]

(i) National wealth
(ii) National income
(iii) Government expenditure
(iv) Money supply

This question is for practice and its solution is available at the end of the chapter

Ans. (i) National wealth is a stock because it is measured at a particular point of time.

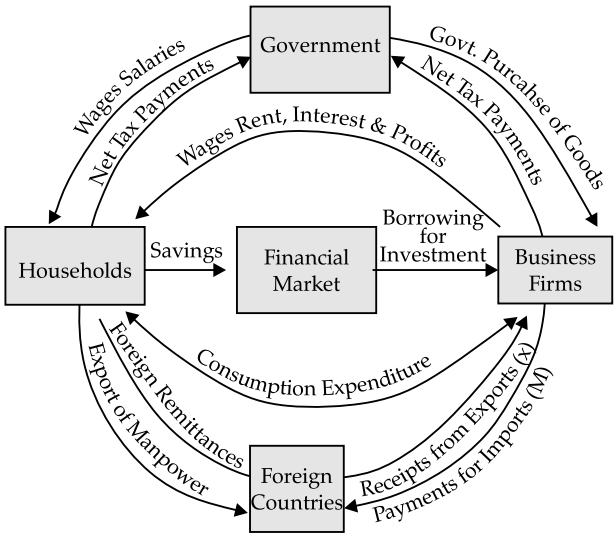
(ii) National income is a flow because it is measured for a period of time.

(iii) Government expenditure is a flow because it is measured for a period of time.

(iv) Money supply is a stock variable because it is measured at a particular point of time.

Q.3. Draw a well labelled diagram to explain briefly, the circular flow of income in a four-sector model. [OEB]

Ans.



Explanation: Four sector model includes household sector, producing sector, government sector and foreign sector. It will study a circular flow income in these sectors including rest of the world i.e., open economy income. Here flows from household sector and producing sector to government sector are in the form of taxes. The income received from the government sector flows to producing and household sector in the form of payments for government purchases of goods and services as well as payment of subsides and transfer payments. Here flows from household sector and producing sector to foreign sector are in the form of exports and imports. The income received from the foreign sector flows to producing and household sector in the form of receipts from exports and foreign remittances. Every payment has a receipt in response of it by which aggregate expenditure of an economy becomes identical to aggregate income and makes this circular flow unending.

Topic-2 Basic Concepts of National Income Accounting

Concepts covered: National Income, GNP, NNP, GDP and NFIA, National Disposable Income

Revision Notes

- **National Income :** National Income can be defined from different points of view:
  - **From the point of view of production :** It is the total monetary value of addition to the value of goods and services produced by residents both within and outside the country during the period of an accounting year.
  - **From the point of view of income :** It is the sum total of factor incomes earned by **normal residents** of a country during the period of an accounting year.
  - **From the point of view of expenditure :** It is the sum total of all expenses incurred by consumers on consumption of final goods and services during the period of an accounting year.

Scan to know more about this topic

National Income and concepts of national income

- **Intermediate Goods :** Goods which are used for further production or for resale in the same year are known as Intermediate Goods.
- **Final Goods :** Goods which are used either for final consumption by the consumers or for investment by the producers are called Final Goods.
- **Domestic Territory :** Domestic or Economic territory is the geographical territory administered by a Government within which persons, goods and capital circulate freely.
- **Normal Residents :** A resident (or a normal resident) of a country is a person or institution who ordinarily resides in a country and whose centre of interest also lies in that country.
- **Transfer Payments :** These are all those unilateral payments corresponding to which there is no value addition in the economy, e.g., gifts, donations, etc.

- **Gross National Product** : It is defined as the total value of all final goods and services produced in a country in a particular year, plus the income which is earned by its citizens who are located abroad and minus the income of non-residents located within the country.
- **Net National Product at Factor Cost (NNP_{FC})** : It is the sum total of factor incomes (rent + interest + profits + wages) earned by normal residents of a country during the period of an accounting year. It is also known as National Income.

$$NNP_{FC} = GNP_{FC} - \text{Depreciation}$$
or
$$NNP_{FC} = NDP_{FC} + NFIA$$

- **Net National Product at Market Price (NNP_{MP}) :** It refers to market value of final goods and services produced during the year inclusive of Net Factor Income from Abroad but exclusive of depreciation.

$$NNP_{MP} = GDP_{MP} - \text{Depreciation} + NFIA$$

- **Gross Domestic Product (GDP) :** It is the total value of all the final goods and services by all the enterprises (both resident and non-resident) within the domestic territory of a country in a particular year. GDP is considered as one of the best indicators of judging the economic performance of a country.
- **Nominal Gross Domestic Product :** It refers to market value of the final goods and services produced within the domestic territory of a country during a financial year, as estimated using the current year prices. It is also called GDP at current price.
- **Real Gross Domestic Product :** It refers to market value of the final goods and services produced within the domestic territory of a country during a financial year, as estimated using the base year prices. It is also called GDP at constant price.
- **Net Factor Income from Abroad (NFIA) :** This is the difference between the income earned from abroad for rendering factor services by the normal residents of the country to the rest of the world and the income paid for the factor services rendered by non-residents in the domestic territory of a country.
- **Factor Income :** These are incomes received by the owners of factors of production for rendering their factor services to the producers.
- **Private Income :** It is the total income from all sources (factor income as well as current transfers) that accrues to the private sector during the period of one year.
- **Personal Income :** It is the income actually received by the individual households from all sources in the form of current transfer payment and factor incomes.
- **Personal Disposable Income :** It is the amount which is actually available to the households and the non-corporate businesses after the deduction of all tax obligations to the government.
- **National Disposable Income (NDI) :** It is defined as the maximum, the country can afford to spend on consumption of goods or services during an accounting year without having to finance its expenditure by disposing of assets or by increasing its liabilities.

  National Disposable Income can be 'Gross' or 'Net'.

  Net National Disposable Income = NNP_{FC} + Net Indirect Taxes + Net Current Transfers from Rest of the World

  Gross National Disposable Income = Net National Disposable Income + Depreciation (current replacement cost)
- **Private Sector :** It refers to all those businesses and enterprises which are owned and controlled by the private individuals.
- **Public Sector :** It refers to all those departmental and non-departmental businesses and enterprises which are owned or controlled by government, For Example:
  - Departmental Post offices, Railways, Tourism Corporation of India, etc.
  - Non-departmental Indian Aviation, Indian Oil, etc.
- **Nominal National Income or National Income at Current Prices :** It is defined as the value of current output at current year prices.
- **Real National Income or National Income at Constant Prices :** It is defined as the value of current output at some base year prices.

$$\text{Real NI} = \frac{\text{Nominal NI}}{\text{Current Price Index Number}} \times 100$$

- **GNP Deflator :** It shows the change in GNP with the change in price levels.
- **Normal Residents :** A resident of a country is a person whose centre of economic interest lies in that country.
- **GDP Deflator :**

$$\text{GDP Deflator} = \frac{\text{GDP at Current Price}}{\text{GDP at Constant Price}} \times 100$$

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Give the meaning of subsidies. [OEB]

Ans. Subsidies can be referred to as the economic assistance provided to the household s and firms so as to increase welfare of the people.

Q.2. State the difference between normal resident and domestic territory of a country? [OEB]

Ans. Normal resident can be referred to as those individuals and institutions who ordinarily resides in a country and their centre of economic interest also lies there. Domestic territory can be referred to as the economic territory administered by a government in which goods, services and capital circulate freely.

Q.3. State the difference between final goods and intermediate goods? [OEB]

Ans. Final goods can be referred to as those goods which are used for either final consumption or for final investment. Intermediate goods can be referred to as those goods which are used for final consumption.

Q.4. Purchase of flour by a housewife is a final good. Why? [OEB]

Ans. The flour purchased by the housewife will not be further processed or resold, thus it is a final good.

Q.5. Machinery purchased by a factory owner is a capital good. Why? [OEB]

Ans. Capital goods are used to help in the production process. As machinery will be used in the production process, it is a capital good.

Q.6. State giving reason with the following statement is true or false: [OEB]

(a) "Transfer income is earned concept whereas factor income is a receipt concept."

Ans. False

Reason: Transfer income is a receipt income whereas factor income is earned concept because transfer income includes gifts, scholarships etc. and factor income includes factor income in the form of rent, wages, interest and profit.

(b) State giving reason with the following statement is true or false:
"The value of final goods is not included in national income whereas that of intermediate goods is included in national income."

Ans. False

Reason: This is because the value of final goods is included in calculation of national income and intermediate goods is not included in the calculation of national income because otherwise it will lead to the problem of double counting.

(c) State giving reason with the following statement is true or false:
"The difference between indirect taxes and subsidies is called as Net factor income from abroad."

Ans. False

Reason: The difference between indirect taxes and subsidies is called as Net indirect taxes.

(d) State giving reason with the following statement is true or false:
"The difference between NNP_{FC} and NNP_{MP} is of consumption of fixed capital."

Ans. False

Reason: As the difference between NNP_{FC} and NNP_{MP} is the net indirect taxes. Net indirect taxes is the difference between indirect taxes and subsidies.

2 Marks Questions

Q.1. What is the difference between Real GDP and Nominal GDP? [ISC Specimen Paper, 2023-24]

Ans. Real GDP is the value of gross domestic product of the country calculated at constant prices. On the other hand, Nominal GDP is the value of the gross domestic product of a country calculated at current prices.

Q.2. Explain two differences between factor income and transfer income. [ISC 2018]

Ans. The difference between factor income and transfer income are:

Factor income		Transfer income	
1.	Received for rendering productive service.	**1.**	Received without rendering any service
2.	Comprises rent, wages, interest and profit.	**2.**	Comprises gifts, subsidies, donations, scholarships, etc.
3.	Earned income	**3.**	Unearned income
4.	Included in national income.	**4.**	Not included in national income.

(Any two) [ISC Marking Scheme, 2018]

Examiner's Comment

- While explaining the difference between factor income and transfer income, several candidates wrote that transfer income is income earned in transferring goods or money, which was incorrect.

Answering Tip

- Clearify the differences between factor income and transfer income.

Q.3. How can gross domestic product at factor cost be obtained from gross national product at market price? [ISC 2015]

Ans. *Domestic Product at Factor Cost* to be obtained from *Gross National Product at Market Price.*

GDP_{FC} from GNP_{MP}

GNP_{MP} – net indirect tax – net factor income earned from abroad = GDP_{FC}

Net indirect tax = Indirect tax – Subsidy

Net factor income earned = Factor income received – Factor income given.

[ISC Marking Scheme 2015]

Examiner's Comment

- Some candidates could not answer this part correctly. They only subtracted NIT and not NFIA from GNPMP.

Answering Tip

- Students must be very clear about the formulas.

Q.4. What is GNP at factor cost? Differentiate it with the National Income. [OEB]

Q.5. How can personal income be calculated from the Private Income? [OEB]

Ans. Personal Income=Private Income–Corporate Savings–Corporate Tax

3 Marks Questions

Q.1. What is meant by economic welfare? Explain how GDP is an indicator of economic welfare.

[ISC 2019]

Ans. Economic Welfare refers to material well-being of people. It refers to the total satisfaction derived from the use of goods and services that can be purchased for money.

GDP is an indicator of economic welfare but per capita income is better. This is because if the increase in GDP is accompanied with a rise in population the per capita availability of goods and services will fall. The per capita income becomes a better indicator. Also one must consider real GDP and not nominal. An increase in money national income may be partly due to increase in prices and partly due to increase quantity of goods and services. Real national income is a better indicator because it eliminates the impact of price changes on national income.

[ISC Marking Scheme, 2019]

Examiner's Comment

- Many candidates did not write the concept of economic welfare. Although, several candidates wrote that GDP was a good indicator, they did not mention that PCI was a better indicator.

Answering Tip

- Practice the concept of welfare and economic welfare with the help of examples.

Q.2. Distinguish between real GDP and nominal GDP. Which of these is a better indicator of economic welfare and why? [ISC 2017]

Ans. Real GDP is the total value of goods and services calculated at 'constant' or base price level.

Nominal GDP is the total value of goods and services calculated at 'current' price level.

Real GDP is a better index of welfare of people. When real GDP rises, flow of goods and service tends to rise., other factors remaining constant. This means, greater availability of goods per person, implying higher level of welfare. Also real GDP facilitates periodic comparison of physical output. **[ISC Marking Scheme, 2017]**

Examiner's Comment

- Majority of candidates answered this part correctly. However, the second part of the question, i.e. 'which of these is a better indicator and why', was not answered well. A number of candidates did not relate Real GDP with base price level and nominal GDP with current price level.

Answering Tip

- Learn the definitions of Real GDP and Nominal GDP, with stress on the words Real and Nominal.

Q.3. Distinguish between private income and personal income. [ISC 2016]

Ans.

Private Income	Personal Income
Private income refers to the total income earned by individuals or households before deduction of personal taxes. It includes all sources of income such as wages, salaries, profits, interest, and rent.	Personal income refers to the total income received by individuals or households after deduction of personal taxes. It includes wages, salaries, dividends, interest, rental income, and government transfers (like social security payments or unemployment benefits), but excludes corporate profits and taxes.
Private income doses not account for personal taxes; it is the gross income earned by individuals or households.	Personal income considers taxes, thus representing the net income available to individuals or households after taxes have been deducted.
It encompasses all sources of income for individuals or households, both earned and unearned, without consideration for taxes.	It includes only those incomes that individuals or households receive directly, adjusted for taxes. It excludes income earned by corporations or other entities.
E.g., – Raw material	E.g., – Machine purchased for the installation in the factory

This question is for practice and its solution is available at the end of the chapter

Q.4. The growth of Gross Domestic Product is not a real indicator of economic welfare. Discuss two reasons to justify the given statement. [ISC 2015]

Ans. It is not a real indicator as it does not tell us about:

(i) Inequality of income and wealth

(ii) Poverty

(iii) Whether present production is increased at the cost of future.

(iv) Whether more of defence items produced.

(Any two can be explained)

[ISC Marking Scheme 2015]

Examiner's Comment

- In many cases, instead of giving reasons to justify the statement, some candidates argued why PCI or real PCI is a better indicator. A number of candidates were confused as to why GDP is not a real indicator of economic welfare.

Answering Tip

- Learn how growth of GDP is not a real indicator of economic welfare if people are not living a comfortable life. Different points can be considered.

Q.5. Differentiate between intermediate good and final good. [OEB]

4 Marks Questions

Q.1. Per capita income is not considered a good indicator of the economic welfare. Explain this statement by giving any two reasons. [ISC 2023]

Ans. Per capita income is not considered a good indicator of economic welfare due to following reasons:

(i) Per capita income can sometimes be misleading because there is unequal distribution of wealth between rich and poor.

(ii) Per capita income does not reflect the standard of living of the people. While standard of living is directly concerned with economic welfare.

Q.2. With the help of appropriate examples, briefly discuss Real GDP and Nominal GDP. [ISC 2023]

Ans. Real GDP: Real gross domestic product is an inflation-adjusted measure of the value of all goods and services produced in an economy. It is expressed in base year prices and referred to as a fixed cost price. It is also known as inflation corrected GDP or constant price GDP. It is a reliable index of economic growth.

Nominal GDP: Nominal gross domestic product (GDP) is GDP given in current prices, without adjustment for inflation. Current price estimates of GDP are obtained by expressing values of all goods and services produced in the current reporting period. It means, the market value of the final production of goods and services within a country in a given period using that year's prices.

For example: Suppose current year's production in a hypothetical economy is 1,000 units at a price of ₹ 500, the GDP at current year price is 1000 × ₹ 500 = ₹ 5,00,000, so the nominal GDP is ₹ 5,00,000. Suppose base year price is ₹ 400, the GDP at base year price is 1000 × ₹ 400 = ₹ 4,00,000. So the Real GDP is ₹ 4,00,000.

Q.3. How will the government spending on child immunisation programme affect GDP and welfare? [OEB]

Ans. As government spends on child immunisation programme, it leads to increase in society's welfare. It is in the sense that if complete immunisation is provided to children of a nation, then it prevents them from the deadly diseases and there will be drastic reductions in the number of child falling ill. This, then checks absenteeism and drop out ratios from school. Complete vaccination also helps to reduce premature deaths. All these factors contribute to superior human capital, which ensures the better utilisation of physical resources. With higher probability of getting employed, there would be greater participation in the production and output will be produced. Hence, it will raise the GDP and welfare of the country in long run.

Q.4. Even the expense of government on popularising yoga can affect the GDP and welfare of the nation." Comment. [OEB]

Ans. GDP is considered as an index of welfare of the people. Welfare means sense of material well being among the people. Many activities in the economy are not evaluated in monetary terms. For example, leisure time activities are not included in GDP but they result in welfare of people in the country. When government incurs expenditure to popularize yoga among the masses, it will result in good health and well being of individuals. This will increase efficiency and lead to higher productivity in future in the country. Thus, in spite of being an activity to which it is difficult to put monetary value, yoga will surely result in better mental and physical health of people in the country resulting in overall welfare of the citizen of the country.

This question is for practice and its solution is available at the end of the chapter

6 Marks Questions

Q.1. "GDP is not considered as a true indicator of economic welfare." Explain. **[OEB]**

Ans. GDP may be a good indicator of economic growth but not of economic welfare or economic development because of:

(i) Externalities: Externalities refer to benefits or harms of an activity caused by a firm or an individual, for which they are not paid or penalised. For example, environmental pollution caused by industrial plants is a negative externality and building a flyover is a positive externality.

(ii) Composition of GDP: Composition of GDP may not be welfare oriented. Even when the level of GDP tends to rise there is no direct increase in the welfare of the masses if GDP has risen owing largely to the increase in the production of defence goods.

(iii) Distribution of GDP: If with every increase in the level of GDP, distribution of GDP is getting more unequal, welfare level of the society may not rise. In this situation, the bulk of the population may have even lesser goods than before (even when the overall level of GDP has tended to rise)

(iv) Non-monetary exchanges: Non-Monetary Exchanges are those activities in an economy which cannot be evaluated in terms of money due to non-availability of data, e.g., domestic services provided by family members at home, barter exchanges, etc. Although these activities contribute to welfare, they are a major cause of understanding of GDP in the economy. Therefore, GDP may not give the true picture of welfare of a country.

Q.2. Differentiate between consumer good and capital good. **[OEB]**

Ans. The difference between consumer good and capital good are:

Sl. No.	Basis for comparison	Consumer Goods	Capital Goods
1.	Meaning	Goods that are used by the end user for consumption are called consumer goods. These are final goods.	Goods that are deployed to produce consumer goods are called capital goods. These are intermediate goods.
2.	Marketing	Business to consumer.	Business to business.
3.	Purpose	Bought for personal consumption.	Bought for making other products.
4.	Buyer	Consumer	Manufacturers
5.	Demand	High	Comparatively less
6.	Price Determination	By suppliers	By companies

Topic-3 Measurement of National Income

Concepts covered: 1. Measurement of National Income, a. Value Added Method, b. Income Method, c. Expenditure Method

Revision Notes

➢ **Measurement of National Income:** In every economy the circular flow of production, income and expenditure remains in operation continuously due to economic activities. Production generates income which creates demand and hence expenditure. In this way, the national income of a country may be measured by three alternative methods. These are :

- In the form of flow of goods and services,
- In the form of income flow,
- In the form of expenditure flow.

Scan to know more about this topic

Measurement of National Income

➢ **Value Added Method or Production Method :** Product Method or Valued Added Method is that method which measures the national income by estimating the contribution of each producing enterprise to production in the domestic territory of the country in an accounting year. For measuring national income by this method, we have to estimate the following components :

- **Net Domestic Product at Market Price (NDP_{MP}) :** Gross Valued Added by [Primary Sector + Secondary Sector + Tertiary Sector] + Depreciation.
- **Net National Product at Factor Cost (NNP_{FC}) or NI :** NNP_{FC} or NI = NDP_{MP} – Indirect Tax + Net Income from Abroad.

➢ **Precautions While Using Value Added Method :**

- The value of intermediate goods should not be included.

- Purchase and sale of second hand goods should not be included.
- Imputed value of self-consumed goods should be included, but self-consumed services should not be included.
- Own account production should be included.
- Commission earned on account of sale and purchases of second hand goods is included.

➢ **Income Method :** It measures national income in terms of payments made in the form of wages, rent, interest and profit to the primary factors of production, *i.e.*, labour, land, capital and enterprise respectively for their productive services in an accounting year.

- **Net Domestic Income or Net Domestic Product at Factor Cost :**

  (1) Compensation to Employees + (2) Operating Surplus + (3) Mixed Income from Self Employment.
- **National Income** = Domestic Income + Net Income from Abroad.

➢ **Precautions While Using Income Method :**

- Income from illegal activities like smuggling, theft, gambling, etc., should not be included.
- Corresponding to production for self-consumption as these are reward for rendering services.
- Brokerage on the sale/purchase of shares and bonds is to be included.
- Income in terms of windfall gains should not be included.
- Transfer earning like old age pensions, unemployment allowances, scholarships, pocket expenses, etc. should not be included.

Scan to know more about this topic

GDP Basics of Expenditure Approach

➢ **Expenditure Method :** By this method, the total sum of expenditures on the purchase of final goods and services produced during an accounting year within an economy is estimated to obtain the value of GDP.

➢ **Final Expenditure:** It is the expenditure on the purchase of final goods and services during an accounting year. It is broadly classified into four categories :

- Private final consumption expenditure
- Government final consumption expenditure
- Investment expenditure
- Net exports i.e., difference between exports and imports during an accounting year

➢ **Computation of National Income** (by expenditure method)

NNP_{FC} = GDP_{MP} – Depreciation + NFIA – Net Indirect Tax

Where, GDP_{MP} = Private Final Consumption Expenditure + Government Final Consumption Expenditure + Gross Domestic Capital Formation + Net Exports (Exports – Imports)

Where, Gross Domestic Capital Formation = Gross Domestic Fixed Capital Formation + Change in Stock (Closing Stock – Opening Stock)

➢ **Precautions While Using Expenditure Method**

- Only final expenditure is to be taken into account to avoid error of double counting.
- Expenditure on second hand goods is not to be included.
- Expenditure on transfer payments by the government is not to be included.
- Imputed value of expenditure on goods produced for self consumption should be taken into account.
- Expenditure on shares and bonds is not to be included in total expenditure.

➢ **Gross Domestic Product at Market Price (GDP_{MP})**: (1) Private Final Consumption Expenditure (C) + (2) Government final Consumption Expenditure (G) + (3) Investment or Gross Capital Formation (I) + Net Exports.

➢ **National Income** = GDP_{MP} – Depreciation – Net Indirect taxes + Net Income from abroad.

Example:

1. **Find National Income from following using expenditure method:**

	Items	(₹ in crores)
(i)	Current transfers from rest of the world	250
(ii)	Net Indirect taxes	500
(iii)	Net Exports	– 125
(iv)	Rent	450
(v)	Private Final Consumption Expenditure	4500
(vi)	Net Domestic Capital Formation	1000
(vii)	Compensation of Employees	2500
(viii)	Net Factor Income from Abroad	– 50
(ix)	Government Final Consumption Expenditure	2000
(x)	Profit	1100
(xi)	Mixed Income of Self Employed	2000
(xii)	Interest	1150

Ans. National Income by Expenditure Method = Private Final Consumption Expenditure + Government Final Consumption Expenditure + Net Domestic Capital Formation + Net Exports + NFIA – NIT

National Income (NNP_{FC}) by Expenditure Method = 4500 + 2000 + 1000 + (–125) + (–50) – 500

= ₹6,825 crores

2. **Calculate Net Domestic Product at Factor Cost by (i) Income Method, and (ii) Production Method from the following data:**

	Items	(₹ in crore)
(i)	Net Value Added at market price by primary sector	4,000
(ii)	Wages and salaries	8,000
(iii)	Social Security Contribution by Employers	400
(iv)	Net value added at market price by the secondary sector	2400
(v)	Corporation Tax	120
(vi)	Retained Earnings of Private Corporations	40
(vii)	Net value added at market price by the tertiary sector	5,600
(viii)	Dividend	240
(ix)	Rent	1200
(x)	Interest	1200
(xi)	Net Indirect Tax	800
(xii)	Social Security Contribution by Employees	320

Ans. **(i) By Income Method:** Net Domestic Product at Factor Cost (NDP_{FC}) = Wages and Salaries + Social Security Contribution by Employers + Corporation Tax + Retained Earning of Private Corporations + Dividend + Rent + Interest

= 8,000 + 400 + 120 + 40 + 240 + 1200 + 1200

= ₹11,200 crore.

(ii) By Production Method: Net Domestic Product at Factor Cost (NDP_{FC}) = Net Value Added by Primary Sector + Net Value Added by Secondary Sector + Net Value Added by Tertiary Sector – Net Indirect Taxes

= 4,000 + 2400 + 5600 – 800

= ₹11,200 crore.

Key Terms

- **Value Added :** Means additions in value by each firm in the process of production at each stage.
- **Double Counting :** Counting the value of the same product more than once in calculation of National Income.
- **Wages and Salaries :** Factor payments received by employees in the form of basic, DA, bonus, commission, HRA, perquisites, etc.
- **Employers' contribution of social security :** Amount deposited by the employer in Provident Fund, insurance schemes, etc. of the employees.
- **Operating Surplus :** Total amount of rent, royalty, interest and profit.
- **Mixed Income of self-employed :** Income of the entrepreneurs who use their own services of land, labour and capital.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. State any two methods of measuring national income? [OEB]

Ans. The two methods of measuring national income are:

(i) Production method

(ii) Income method

Q.2. Define net Domestic Product at market price? [OEB]

Ans. Net domestic product at market price can be referred to as the net market value of all final goods and services produced within the domestic territory of a country.

Q.3. Give the meaning of national income? [OEB]

Ans. National income can be referred to as net money value of all final goods and services produced by the normal residents of a country during a year.

Q.4. Define mixed income. [OEB]

Ans. Mixed income can be referred to as the income generated by own account workers such as farmers, lawyers etc whose income cannot be classified in more than one type of factor income.

Q.5. State giving reason when net exports can be negative? [OEB]

Ans. Net exports can be negative when imports is greater than exports.

Q.6. State whether the following statement is true or false: [OEB]

(a) "Value added is equal to value of output plus intermediate consumption."

Ans. False

Reason: Because value added is equal to value of output minus intermediate consumption.

(b) "Goods purchase for self-consumption are not a part of national income."

Ans. False

Reason: Goods purchased for self-consumption are a part of national income because these are part of current production.

(c) "Undistributed profit, corporate tax and dividend are part of profit."

Ans. True

Reason: Profit has three components such as undistributed profit, corporate tax and dividend.

(d) "Net Domestic capital formation comprises of net domestic fixed capital formation and consumption of fixed capital."

Ans. False

Reason: Net domestic capital formation comprises of net domestic fixed capital formation, consumption of fixed capital and change in stock.

(e) "Sale of second hand goods is included in national income."

Ans. False

Reason: Sale of second hand goods is not included in national income because second hand goods are not part of current production.

(f) "Sale and purchase of shares is a part of national income."

Ans. False

Reason: This is because sale and purchase of shares is not a part of national income because it does not contribute to the current production.

2 Marks Questions

Q.1. Why are transfer incomes and windfall gains not included in the calculation of national income? [ISC Specimen Paper, 2023-24]

Ans. Transfer incomes like scholarships, donations, charity, old age pensions, etc. are not included in the National income because such receipts are not connected with any productive activity and there is no value addition.

Windfall gains like income from lotteries, horse race, etc. are not included as there is no productive activity connected with them.

Q.2. Briefly discuss any two precautions to be taken while calculating national income by the value – added method. [ISC 2023]

Concept Applied	Precautions While Using Value Added Method

Ans. Precautions while using Value Added Method:

(i) Intermediate Goods are not to be included in the national income since such goods are already included in the value of final goods. If they are included again, it will lead to double counting.

(ii) Sale and purchase of second-hand goods is not included as they were included in the year in which they were produced and do not add to current flow of goods and services. However, any commission or brokerage on sale or purchase of such goods will be included in the national income as it is a productive service.

(iii) Production of services for self-consumption or domestic services are not included. Domestic services like services of a housewife, kitchen gardening, etc. are not included in the national income since it is difficult to measure their market value. These services are produced and consumed at home and never enter the market place and are termed as non-market transactions.

(iv) Production of goods for self-consumption will be included in the national income as they contribute to the current output. Their value is to be estimated or imputed as they are not sold in the market.

(v) Imputed value of owner-occupied houses should be included. People who live in their own houses, do not pay any rent. But, they enjoy housing services similar to those people who stay in rented houses. Therefore, value of such housing services is estimated according to market rent of similar accommodation. Such an estimated rent is known as imputed rent.

(vi) Change in stock of goods will be included. Net increase in the stock of inventories will be included in the national income as it is a part of capital formation. **[Any Two]**

Q.3. State the components of compensation of employees. Give one example of any one of the components. [ISC Semester-II, 2022]

OR

Explain the components of compensation of employees for calculation of National Income by Income method. [ISC 2018]

Ans. Compensation of Employees (COES): COE refers to amount paid to employees by employer for rendering productive services. It includes all payments and benefits, which the employees receive, directly or indirectly from the employer.

Compensation of employees consists of the following components:

(i) Wages and salaries in cash: It includes all monetary benefits, like wages, salaries, bonus, dearness allowances, commission, etc.

(ii) Wages and salaries in kind: It includes all non-monetary benefits, like rent-free home, free car, free medical and educational facilities, etc.

(iii) Employer's contribution to social security schemes: It includes contributions made by employer for the social security scheme of employees. For example, contribution to provident fund, gratuity, labour welfare funds, retirement pension, etc.

Examiner's Comment

- Compensation of employees for calculation of National Income under the three components- wages and salaries in cash, in kind and supplementary labour incomes, was not explained by many candidates. Several candidates did not mention these three components or wrote different items under them in groups or placed items of one component under another one. A large number of candidates were confused between items of wages in kind and supplementary income.

Answering Tip

- TIn order to avoid rote learning, the students need to analyse the reason or logic behind the inclusion of various items under the specific components of compensation of employees.

Q.4. State whether the following will be included or not in the national income accounting. Give one reason for each. [ISC Semester-II, 2022]

(i) Interest paid on a loan taken to buy consumer durable good.

(ii) Food grain grown by a farmer for self consumption.

Ans. (i) Yes, it is included in national income accounting as it is a part of consumption expenditure.

(ii) Yes, it is included in national income accounting as goods have a market value. Grains grown by a farmer but used entirely for self-consumption is a part of the contribution by him to the value of goods produced. Therefore, this is included in national income.

Q.5. What is meant by operating surplus? [ISC 2020]

Ans. Operating surplus is the income from property and entrepreneurship

OR

It is the income earned from the ownership and control of capital. (Rent + Interest + Royalties +dividends + undistributed profits before corporation taxes + profits of Government ent.)

[ISC Marking Scheme, 2020]

Examiner's Comment

- Most candidates did not write entrepreneurship in their answers. Several candidates did not mention the components of operating surplus.

Answering Tip

- At the start of any topic, ensure that the meaning of the different terms is properly understood and ensure that the meaning and significance of each and every term in the definition is clear. Concepts like, what is the income from property and entrepreneurship should be emphasised upon. The components of operating surplus should be written on and learnt as per the number of items coming under them.

Q.6. Explain any two precautions to be taken while calculating national income by income method. [ISC 2019]

Concept Applied	Precautions while using income method

Ans. Precautions to be taken while calculating national income by income method:

(i) Transfer incomes like scholarships, donations, charity, old age pensions, etc. are not included in the National income because such receipts are not connected with any productive activity and there is no value addition.

(ii) Income from sale of second-hand goods will not be included in national income as their original sale has already been counted. If they are included again, it would lead to double counting. However, any brokerage or commission received by brokers or commission agents on sale of such goods will be included as it is an income received for rendering productive service.

(iii) Income from sale of shares, bonds and debentures will not be included as such transactions do not contribute to current flow of goods and services. These financial assets are mere paper claims and involve a change of title only. However, any commission or brokerage on such financial assets is included as it is a productive service.

(iv) Windfall gains like income from lotteries, horse race, etc. are not included as there is no productive activity connected with them.

(v) Imputed value of services provided by owners of production units will be included. Imputed value of owner-occupied houses, interest on own capital, production for self-consumption, etc. will be included as these are productive activities and add to the flow of goods and services.

(vi) Payments out of past savings like death duties, gift tax, wealth tax, etc. are not included in the national income because they are paid out of wealth or past savings and do not add to current flow of goods and services. Indirect taxes are not included in national income at factor cost. However, they are included in national income at market price. **[Any Two]**

Examiner's Comment

- Majority of the candidates wrote the points for calculating national income without mentioning whether they should be included or excluded. Many candidates also included the precautions of other methods like output method and expenditure method which were not asked for.

Answering Tip

- Learn the precautions of each method of measuring National Income separately, laying stress on the fact that precautions are not common to all the methods.

Q.7. Explain with the help of an example, the problem of double counting while calculating national income. [ISC 2017]

Ans. The problem of double counting is the problem of estimating the value of goods and services more than once. While estimating national income by using the output method, only the value of final goods and services is taken into consideration. This leads to overestimation of the value of goods and services produced.

[ISC Marking Scheme, 2017]

Examiner's Comment

- In many answers, inclusion of second-hand goods and pure exchange transactions were considered as double counting, which was incorrect.

Answering Tip

- The students need to understand that double counting problem refers to the output method.

Q.8. State whether each of the items given below is included in estimating national income. Give a reason in each case to justify your answer.

(i) Expenditure on the construction of express highway

(ii) Expenditure on the purchase of an old house

[ISC 2016]

Ans. (i) Yes, because it is a part of gross fixed investment

(ii) No, because old house is not a part of current production. **[ISC Marking Scheme 2016]**

3 Marks Questions

Q.1. How can you obtain: [ISC 2020]

(i) National income from GDP_{MP}.

(ii) Personal disposable income from personal income.

Ans. (i) National Income from GDPMP:

$NNP_{FC} = GDP_{MP}$ – depreciation – NIT + NFIA

(ii) Personal disposable income from personal income:

Personal disposable income = Personal income personal tax paid by household other miscellaneous receipts of the government administrative departments.

[ISC Marking Scheme. 2020]

Examiner's Comment

- Most of the candidates got confused with the items to be deleted and added for calculating National Income from GDPMP. To derive Personal Disposable Income from Personal Income, many candidates did not subtract miscellaneous receipts of Government administrative departments. Some candidates wrote only miscellaneous receipts. A few candidates obtained the required concepts in the reverse order.

Answering Tip

- Understand the meaning of the phrase, 'other miscellaneous receipts of the government', with the help of examples such that its relevance is understood and are thus able to avoid making mistakes.

Q.2. Explain how the following are treated in estimating National Income: [ISC 2018]

(i) Wheat grown by a farmer for self-consumption.

(ii) Earnings of the shareholders from the sale of shares.

(iii) Services rendered by family members to each other.

Ans. (i) Wheat grown by a farmer for self-consumption. To be counted as it is production in that year. It is added to the national income.

(ii) Earnings of the shareholders from the sale of shares are excluded as they only reflect the transfer of an asset, not an addition to the National Income.

(iii) Services rendered by family members to each other. Excluded as there is no way of evaluating these services. **[ISC Marking Scheme, 2018]**

Examiner's Comments

- Several candidates wrote that wheat grown by a farmer for self-consumption should not be included in estimating National Income.
- A few candidates wrote that the earning of the shareholders should not be included in estimating National Income but did not write the reason for the same.
- Several candidates wrote the correct answer that services rendered by family members to each other should be excluded in estimating National Income but were unable to write the reason for the same.

Answering Tips

- The students should know that wheat grown by the farmer is included in the calculation of National income as it is produced during the year.
- The students should analyse that earning of the shareholders from the sale of shares is excluded in estimating national income as it is merely transfer of funds.
- The students should emphasise on the fact that services rendered by family members to each other are excluded as services cannot be expressed in monetary terms.

Q.3. Explain the following components of domestic factor income.

(i) Compensation of employees

(ii) Operating Surplus **[ISC 2016]**

Ans. (i) Compensation of employees: Payment made to labourers in terms of wages, and salaries in cash, in kind and contribution to social security by the employers.

(ii) Operating surplus: It is the income earned from property and entrepreneurship, rent, interest, profit, etc. **[ISC Marking Scheme, 2016]**

Examiner's Comments

- All the three aspects – wages and salaries in kind, wages and salaries in cash and employer's contribution to social security were not written by a number of candidates.
- In most of the answers given by candidates, operating surplus was not defined correctly. Candidates only mentioned the components as rent, interest, profit.

Answering Tip

- Students must need to clear their concepts to avoid mistakes.

Q.4. How can you obtain: **[OEB]**

(i) National Income from GDP_{MP}.

(ii) Personal disposable income from personal income.

Q.5. Explain the steps involved in calculating the National income by Income method. **[OEB]**

Ans. Steps of Income Method Formula:

(i) Identification and Classification of Production Units

The first step in calculating national income via income method is to identify and segregate the units of production. They are classified into three categories, primary, secondary and tertiary.

(ii) Classify and Estimate the Factor Income

The next step of the income method of national income is to classify factor payments in different categories like wages, rent, interest, profit and mixed income. Otherwise, they can be classified into compensation to employees, operating surplus and mixed-income.

After classifying, estimate the number of such payments made by enterprises.

(iii) Calculating Domestic Income

Summing up all factor incomes of every sector will present the domestic income figure (NDPFC).

NDPFC = Compensation of Employees + Operating Surplus + Mixed-Income

(iv) Estimate NFIA to arrive at National Income

The last step to reach the final National Income figure is to estimate Net Factor Income from Abroad (NFIA) with NDP_{FC}.

National Income (NNP_{FC}) = Net Domestic Product at Factor Cost (NDP_{FC}) + Net Factor Income from Abroad (NFIA)

4 Marks Questions

Q.1. Explain the steps involved in calculation of National Income by Expenditure Method.

[ISC Semester-II, 2022]

Ans. The steps involved in calculating National Income by Expenditure Method are:

Step 1: Identify the Sectors Incurring Final Expenditure: All sectors which incur final expenditure within the domestic territory are classified into four groups: (i) Household sector; (ii) Government sector; (iii) Producing sector; (iv) Foreign sector.

Step 2: Classification of Final Expenditure: Final expenditures incurred by the above-mentioned sectors are estimated and classified under the following heads:

(i) Private Final Consumption Expenditure (PFCE)

(ii) Government Final Consumption Expenditure (GFCE)

(iii) Gross Domestic Capital Formation (GDCF)

(iv) Net Exports (X – M).

The sum of four components of final expenditure gives Gross Domestic Product at Market Price (GDP_{MP}), i.e. GDP_{MP} = PFCE + GFCE + GDCF + (X-M)

Step 3: Calculation of Domestic Income (NDP_{FC}): By subtracting the amount of depreciation and net indirect taxes from GDP_{MP}, we get domestic income, i.e., NDP_{FC} = GDPMP – Depreciation – Net Indirect Taxes.

Step 4: Estimate Net Factor Income from Abroad (NFIA) to arrive at National Income: In the final step, NFIA is added to domestic income to arrive at National Income. National Income (NNPFC) = NDPFC+ NFIA

Q.2. Calculate: (i) Operating surplus, (ii) Domestic Income. **[OEB]**

S. No.	Components	(₹ in crores)
1	Compensation of employees	4000
2	Rent and Interest	1600
3	Indirect taxes	240
4	Corporation tax	920
5	Depreciation	200
6	Dividend	1880
7	Mixed Income	400

This question is for practice and its solution is available at the end of the chapter

8	Subsidies	40
9	Undistributed profits	600
10	Net Factor Income to abroad	300

Ans. (i) operating surplus = Rent and interest + Corporation tax+ Dividend+ Undistributed profits = 1600 + 920 + 1880 + 600 = ₹ 5000 crores

(ii) NDP_{FC} = Operating surplus + compensation of employees + mixed-income of self employed = 5000 + 4000 + 400 = ₹ 9400 crores

Q.3. Find out: (i) Net National Product at Market Price, and (ii) Gross Domestic Product at Factor Cost: [OEB]

S. No.	Components	(₹ in crores)
1	Rent and Interest	12000
2	Wages and Salaries	3600
3	Net Indirect taxes	200
4	Corporation tax	240
5	Net Factor Income to abroad	140
6	Mixed Income	2000
7	Social Security Contribution by employers	400
8	Depreciation	100
9	Dividend	160
10	Undistributed profit	800
11	Subsidies	40

Ans. (i) Net National Product at Market Price = Rent and interest +Dividend + Undistributed profit + corporation tax + wages and salaries + Social security contribution by employers + Mixed Income = 12000 + 160 +800 + 240 + 3600 + 400 + 2000 = ₹ 19200 crores

(ii) Gross domestic product at factor cost = NNP_{MP} + Depreciation + Net indirect taxes = 19200 +100 + 200 = ₹ 19500 crores.

Q.4. Find out: (i) National Income, and (ii) Domestic income [OEB]

S. No.	Components	(₹ in crores)
1	Mixed income of self-employed	30,000
2	Dividends	400
3	Interest	800
4	Profit	2000
5	Depreciation	800
6	Compensation of employees	14000
7	Net factor income to abroad	200
8	Exports	200
9	Imports	400
10	Indirect taxes	1600
11	Rent	1000

Ans. National income(NNPFC) = Rent + Interest + Profit + Compensation of employees + Mixed income of self employed = 1000 + 800 + 2000 + 14000 + 30,000 = ₹ 47800 crores.

Domestic income = National income – Net factor factor income to abroad

= 47800 – (-200)

= ₹ 48000 crores

Q.5. Find out Gross National Product at Market Price: [OEB]

S. No.	Components	(₹ in crores)
1	Wages and salaries	1600
2	Undistributed Profit	20
3	Operating Surplus	400
4	Corporate tax	100
5	Social Security Contribution by employers	200
6	Net Factor income to abroad	-40
7	Net Indirect tax	140
8	Depreciation	60
9	Mixed income of self-employed	1000
10	Royalty	18

Ans. GNP_{MP} = NDP_{FC} + depreciation + net indirect taxes + net factor income from abroad

= operating surplus + wages and salaries + social security contribution by employers + mixed income of self-employed + depreciation + net indirect taxes + net factor income from abroad

= 400 + 1600 + 200 + 1000 + 60 + 140 – (-40)

= ₹ 3440 crores

6 Marks Questions

Q.1. Calculate National Income by Income Method and Expenditure Method from the following information: [ISC Specimen Paper 2023-24]

	Items	₹ (in Crore)
1.	**Government final Consumption Expenditure**	**1,100**
2.	**Net Exports**	**(-) 30**
3.	**Mixed income of self employed**	**700**
4.	**Interest**	**620**
5.	**Rent**	**400**

6.	**Consumption of fixed capital**	**130**
7.	**Net domestic capital formation**	**770**
8.	**Private final consumption expenditure**	**2000**
9.	**Profit**	**800**
10.	**Net indirect taxes**	**120**
11.	**Net factor income from abroad**	**(–) 20**
12.	**Compensation of employees**	**1200**

Concept Applied: Calculation of National Income by Income and Expenditure Method

Ans. By Income Method:

NNP_{FC} = Mixed Income of self-employed (3) + Interest (4) + Rent (5) + Profit (9) + Compensation of Employees (12) + Net factor income from abroad (11).

= 700 + 620 + 400 + 800 + 1200 + (–) 20 = ₹3,700 crore.

By Expenditure Method:

NNP_{FC} = Government final consumption expenditure (1) + Private final consumption expenditure (8) + Net Domestic Capital Formation (7) + Net Exports (2) – Net Indirect tax (10) + Net factor income from abroad (11)

= 1,100 + 770 + 2,000 + (-) 30 – 120 + (–) 20 = ₹3,700 crore.

Q.2. Calculate GDP_{MP} and NNP_{MP} from the following data: [ISC Specimen Paper, 2023-24]

	Items	₹ (in Crore)
1.	**Private final consumption expenditure**	**1,000**
2.	**Net domestic capital formation**	**200**
3.	**Government final consumption expenditure**	**500**
4.	**Consumption of fixed capital**	**60**
5.	**Net factor income from abroad**	**(–) 10**
6.	**Indirect tax**	**100**
7.	**Net exports**	**(–) 20**
8.	**Subsidies**	**20**

Ans. GDP_{MP} = Private final consumption expenditure (1) + Net domestic Capital formation (2) + Consumption of fixed capital + Government final consumption expenditure (3) + Net Exports (7)

= 1,000 + 200 + 60 + 500 + (-) 20 = ₹1,740 crore

NNP_{MP} = GDP_{MP} – Indirect tax (6) + Subsidies (8) + Net factor income from abroad (5)

= 1,740 – 100 + 20 + (-) 10 = ₹1,650 crore

Q.3. Calculate GDP_{MP} and NNP_{FC} by Value Added method from the following data. [ISC 2023]

	Particulars	₹ (crores)
(i)	**Net value added at factor cost in Primary sector**	**6000**
(ii)	**Net value added at factor cost in Secondary sector**	**4000**
(iii)	**Net value added at factor cost in Tertiary sector**	**4500**
(iv)	**Net Factor Income from Abroad**	**(–)50**
(v)	**Net Indirect taxes**	**150**
(vi)	**Intermediate consumption**	**2500**
(vii)	**Depreciation**	**500**

Concept Applied: Calculation of National Income by Value Added Method

Ans. Estimation of Gross Domestic Product at Market Price or GDP_{MP} by Value added Method

GDP_{MP} = Net Value Added at factor cost in Primary Sector + Net Value Added at factor cost in Secondary Sector + Net Value Added at factor cost in Tertiary Sector + Net Indirect Tax + Depreciation

= 6000 + 4000 + 4500 + 150 + 500

= ₹ 15150 crore

NNP_{FC} = Net Value Added at factor cost in Primary Sector + Net Value Added at factor cost in Secondary Sector + Net Value Added at factor cost in Tertiary Sector + Net factor income from Abroad

= 6000 + 4000 + 4500 + (–)50

= ₹ 14450 crore

Q.4. Calculate National Income and Operating Surplus from the following data: [ISC 2023]

	Particulars	₹ (crores)
(i)	**Government final consumption expenditure**	**900**
(ii)	**Net factor income from abroad**	**210**
(iii)	**Private final consumption expenditure**	**1000**
(iv)	**Net domestic capital formation**	**300**
(v)	**Profits**	**320**
(vi)	**Rent**	**190**
(vii)	**Net exports**	**(–)75**
(viii)	**Interest**	**200**
(ix)	**Net indirect taxes**	**265**

Ans. Measure of National Income by Expenditure Method are as follows:

National Income = Governmental Final Consumption Expenditure + Private Final Consumption Expenditure + Net Domestic Capital Formation + Net Factor Income from Abroad + Net Export – Net Indirect Tax

NNP_{FC} = 900 + 1000 + 300 + 210 + (–)75 – 265

= ₹ 2070 crores

Operating Surplus = Rent + Profit + Interest

= 190 + 320 + 200

= ₹ 710 crores

Q.5. From the following data, calculate GNP_{FC}, NDP_{FC} and national income: [ISC 2020]

	Item	₹ in crores
(i)	Private final consumption expenditure	950
(ii)	Gross domestic fixed capital formation	370
(iii)	Consumption of fixed capital	20
(iv)	Government final consumption expenditure	410
(v)	Closing stock	300
(vi)	Subsidy	80
(vii)	Net exports	(–)50
(viii)	Wages and salaries	780
(ix)	Net factor income from abroad	(–)40
(x)	Indirect tax	180
(xi)	Opening stock	150
(xii)	Profit before tax	200

Ans. GNPFC = GDPMP + NFIA – NIT
= 1830 + ix – (x-vi)
= 1830 + (-40) – (180-80)
= ₹ 1690 Crores
NDPFC = GDPMP – depreciation – NIT
= 1830 – iii – (x-vi)
= 1830 – 20 – (180-80)
= ₹ 1710 Crores
NNPFC = NDPFC + NFIA = 1710 + ix
= 1710 + (-40) = ₹1670 Crores

[ISC Marking Scheme, 2020]

Examiner's Comment

- Majority of the candidates calculated GNPFC, NDPFC and NNPFC or NI on the basis of GDPMP. While calculating GDPMP,, many candidates did not consider Change in stock, i.e., the difference between closing stock and opening stock (Item no. v - item no. xi), thereby getting an incorrect GDPMP. This had a cascading effect as due to the substitution of incorrect value of GDPMP, all other variables were also incorrect.

Answering Tips

- Emphasise that all the formulae and their derivations have to be learnt.
- Give sufficient practice in solving numericals on National Income, preferably with various components, as any typical component if treated incorrectly can result in the entire numerical being incorrect. The students should practice to solve numericals step by step and stress upon writing units.

Q.6. From the following data, calculate National Income by Output method and Income method: [ISC 2019]

	Item	₹ in crores
(i)	Value of output	2500
(ii)	Value of intermediate consumption	1300
(iii)	Subsidies	40
(iv)	Rent	110
(v)	Employer's contribution to social security	30
(vi)	Profit	50
(vii)	Wages and salaries	340
(viii)	Interest 10	
(ix)	Mixed income of self-employed	360
(x)	Indirect tax	180
(xi)	NFIA	(–)30
(xii)	Consumption of fixed capital	160

Ans. (i) Value added / output Method:
NVAFC = value of output – Intermediate consumption – depreciation – Net indirect tax
NVA_{FC} = 2500 – 1300 – 160 –(180-40) = ₹ 900 cr.
NNP_{FC} = 900 + (–30) = ₹ 870 Cr.

(ii) Income Method:
NDPFC = COE + OS + MI
COE = wages + salaries + employer's contribution to social security
COE = 340 + 30 = 370
OS = R + I + Profit
OS = 110 + 10 + 50 = 170
MI = 360
NDP_{FC} = 370 + 170 + 360 = ₹ 900 cr.
NNPFC = NDPFC + NFIA
= 900 + (–30) = ₹ 870 cr.

[ISC Marking Scheme, 2019]

Examiner's Comment

- Several candidates were unable to treat correctly the item NFIA in both output method and income method. Many candidates were confused with GDP, NDP and NNP at factor cost. Some candidates tried solving the problem in one line and hence could not get the correct answer.

Answering Tips

- Give adequate practice to solving problems on National Income so that they become familiar with the terminology and treatment of the various items. Emphasise on the use of NFIA.
- The students are advised not to miss steps while solving the sums.
- The students must write the formula and then substitute the values in it.

Q.7. From the following data, calculate National Income by Income method and Expenditure method: [ISC 2018]

	Item	₹ in crores
(i)	Compensation of employees	700
(ii)	Government final consumption expenditure	750
(iii)	Net factor income from abroad	(–)10
(iv)	Net exports	(–)15
(v)	Profits	600
(vi)	Net indirect taxes	60
(vii)	Mixed income of self employed	350
(viii)	Rent	200
(ix)	Interest	310
(x)	Private final consumption expenditure	1100
(xi)	Net domestic capital formation	385
(xii)	Consumption of fixed capital	65

Ans. Income Method:

Compensation of employees + Profits + Rent + Interest + Mixed income of self- employed + NFIA.

700 + 600 + 200 + 310 +350+(– 10) = ₹ 2150 Cr.

Expenditure method:

Private final consumption expenditure + Net domestic capital formation + Govt. final consumption expenditure + net exports – NIT + NFIA

1100 + 385+ 750 + (−15) – 60 + (−10) = ₹ 2150 Cr.

[ISC Marking Scheme, 2018]

Examiner's Comment

- Many candidates calculated the incorrect National Income as the formula used by them under the Income method was GDPFC = C.E. + profit + rent + interest + mixed income of self-employed. Under the Expenditure method GDPMP was incorrectly calculated by including net domestic capital formation along with other items.

Answering Tips

- Practice sufficient number of sums so that confidence can be gained.

Q.8. Calculate GNP at FC from the following data by using income method and expenditure method: [ISC 2017]

	Item	₹ in crores
(i)	Operating surplus	600
(ii)	Exports	30
(iii)	Imports	60
(iv)	Private final consumption expenditure	1000
(v)	Net indirect tax	60
(vi)	Compensation of employees	900
(vii)	Mixed income of self employed	160
(viii)	Gross domestic capital formations	330
(ix)	Depreciation	30
(x)	Net factor income from abroad	(–20)
(xi)	Govt. final consumption expenditure	450

Ans. By Income Method:

NDPFC= [(vii) + (i) + (viii)] = 900+600+160=1660 Crores

GNP_{FC}= NDPFC+ Depreciation+ Net factor income from abroad

= 1660+30+ (-20) = 1670 Crores

By Expenditure Method:

GDPMP= [(v) + (iv) + (ix) + (ii)-(iii)] = 450 + 1000 + 330 + 30 – 60 =1750 Crores

GNP_{FC}= GDP_{MP} +Net factor Income from abroad – Net Indirect tax

=1750 + (–20) – 60 =1670 Crores

[ISC Marking Scheme, 2017]

Examiner's Comment

- Many candidates committed mistakes in the numerical on National Income. A number of candidates calculated NDPfc using income method by adding operating surplus, compensation of employees and mixed income but wrote GNPfc.

Answering Tips

- Solve questions on national income step wise with method to convert – national to domestic, market price to factor price, gross to net, etc.
- Clearly learn with examples the distinction between NDP, NNP, GDP, GNP, NNPmp and NNPfc.

Q.9. From the following data, calculate GNP_{MP} and National Income by using value added method:

Gross value of output in primary sector (at FC)	**950**
Gross value of output in secondary sector (at FC)	**470**
Gross value of output in tertiary sector (at FC)	**500**
Value of intermediate goods in primary sector	**360**
Value of intermediate goods in secondary sector	**200**
Value of intermediate goods in tertiary sector	**175**
Depreciation	**20**
Indirect taxes	**35**
Subsidy	**10**
Net factor income from abroad	**4**

[ISC 2016]

Ans. GDP_{FC} = Gross Value of Output in primary sector + Gross Value of Output in secondary sector + Gross Value of Output in tertiary sector – Value of intermediate consumption in primary sector – Value of intermediate consumption in secondary sector – Value of intermediate consumption in tertiary sector

$GNP_{MP} = GDP_{FC}$ + NFIA

1185 + 4

= 1189 crore

National Income = GNP at MP – Dept – NIT

OR

NNP_{FC} = 1189 – 20 – 25

= 1144 crore

[ISC Marking Scheme 2016]

Examiner's Comment

- A number of candidates made mistakes in calculation. The question was already given at factor cost which a number of candidates overlooked.

Answering Tip

- Students must regularly practice numericals to avoid calculation mistake.

Q.10. From the following data calculate GNP_{MP} and NNP_{FC} by expenditure method:

Government final consumption expenditure	**550**
Net factor income earned from abroad	**(–) 20**
Private final consumption expenditure	**1000**
Net domestic capital formation	**380**
Profits	**400**
Rent	**150**
Net exports	**(–) 30**
Interest	**200**
Net indirect taxes	**150**
Mixed Income of Self-employed	**450**
Compensation of employees	**550**
Consumption of Fixed Capital	**170**

[ISC 2015]

Ans. GNP_{MP} = Private final consumption expenditure + Net domestic capital formation + Government Final Consumption Expenditure + Net exports + Net factor income from abroad + depreciation

= 1000 + 380 + 550 + (–30) + (–20) + 170 = 2050

$NNP_{FC} = GNP_{MP}$ – Depreciation – Net Indirect taxes = 2050 – 170 – 150 = 1730

[ISC Marking Scheme 2015]

Examiner's Comment

- The 'Net' of 'Net domestic capital formation was overlooked by a number of candidates. It was treated as 'gross' instead. Hence, instead of NDP_{MP} candidates calculated GDP_{MP} In some cases, Income Method was used to calculate GNP_{MP} and NNP_{FC} though it was clearly mentioned in the question that the expenditure method was to be used.

Answering Tip

- The formulas should be on the tips of the students.

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. Which one of the following best illustrates leakages in the circular flow of income?

[ISC Specimen Paper, 2023-24]

(A) Investment **(B)** Export earning
(C) Savings by households
(D) Government expenditure

Ans. Option (C) is correct.

Q.2. Which one of the following is NOT an example of leakage in the circular flow of income?

[ISC Semester-II, 2022]

(A) Taxation **(B)** Import
(C) Investment **(D)** Saving

Ans. Option (C) is correct.

Explanation: Leakage occurs when there is a withdrawal of money from the economy that results in a reduction of the national income. Sources of

leakages include taxes, savings and imports. But investment is an example of injections in circular flow of income.

Q.3. Read the following Statements 1 and 2 and choose the correct alternative:
Statement 1: Bread purchased by a housewife for making sandwiches is a final good.
Statement 2: Cement purchased by a household to build his house is an intermediate good.
(A) Both the statements are true.
(B) Both the statements are false.
(C) Only Statement 1 is true.
(D) Only Statement 2 is true. **[OEB]**

Ans. Option (C) is correct.
Explanation: Cement purchased by the household to build a house is a final good as it is not purchased for resale but is purchased to build his own house.

Q.4. Read the following Statement 1 and 2 and choose the correct alternatives:
Statement 1: Households sell labour and get wages in return.
Statement 2: Firms sell products and get profits in return.
(A) Both the statements are true.
(B) Both the statements are false.
(C) Only Statement 1 is true.
(D) Only Statement 2 is true. **[OEB]**

Ans. Option (A) is correct.
Explanation: Circular Flow of income in a two-sector economy: Households are owners of factor of production, they provide factor services to the firms (producing units). Firms provide factor payments in exchange of their factor services. So, factor payments flow from firms (producing units) to households. Households purchase goods and services from firms (producing units) for which they make payment to them. So, consumption expenditure (spending on goods and services) flows from households to the firms.

Q.5. Identify the correctly matched pair of items in Column A to those in Column B: [OEB]

Column A		Column B	
1.	Income Tax	(a)	Forced Transfer
2.	Services of Housewives	(b)	Market Activities
3.	Retirement Pension	(c)	Taxable for Firm
4.	Annual value of goods and services produced	(d)	Income method

(A) 1 – (a) **(B)** 2 – (b)
(C) 3 – (c) **(D)** 4 – (d)

Ans. Option (A) is correct.
Explanation: Income tax is a forced transfer as taxpayer cannot avoid it.

Q.6. Read the following statements 1 and 2 and choose the correct alternatives:
Statement 1: Payment of interest by a government firm is included in the National Income.
Statement 2: Festival gifts to employees is not included in the National Income.
(A) Both Statements are true.
(B) Both Statements are false.
(C) Only Statement 1 is true .
(D) Only Statement 2 is true. **[OEB]**

Ans. Option (A) is correct.
Explanation: Payment of interest by a government firm should be included while estimating National Income because it is a kind of factor payment. Festival gift from an employer is not included in the estimation of National Income because it is a transfer payment.

Q.7. With a rise in real national income, welfare of the people:
(A) rises **(B)** falls
(C) remains unchanged **(D)** none of the above **[OEB]**

Ans. Option (A) is correct.
Explanation: With a rise in real national income, welfare of the people rises as people have more money to spend on the needs and wants.

Q.8. GDP is not an appropriate indicator of welfare because of: [OEB]
(A) externalities.
(B) composition of GDP and distribution of GDP.
(C) non-monetary transactions.
(D) all of the above

Ans. Option (D) is correct.
Explanation: There are certain limitations of GDP as an index of welfare:
(i) GDP does not consider changes in inequalities in the distribution of income.
(ii) GDP does not include non-monetary exchanges.
(iii) GDP does not consider the changes in the population of a country.
(iv) GDP does not account for valuation of externalities.

Q.9. Pollution created by factories/vehicles is an example of: [OEB]
(A) positive externalities.
(B) negative externalities.
(C) either (A) or (B).
(D) neither (A) of (B).

Ans. Option (B) is correct.
Explanation: Cost suffered by unrelated third party due to result of an economic transaction is known as negative externalities. Pollution created by factories/ vehicles has negative effect on the society.

Q.10. Real GDP is considered as an index of: [OEB]
(A) price level in the economy.
(B) welfare of the people.
(C) profit maximisation.
(D) none of the above.

Ans. Option (B) is correct.
Explanation: Real GDP is considered as an index of welfare of the people because real GDP measures the goods and services without calculating the impact of price level in the economy.

[B] Assertion Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.1. Assertion (A): The flour purchased by a baker is considered an intermediate good.
Reason (R): The flour is used for the production of bread, cakes and biscuits to be sold to the consumer. **[OEB]**

Ans. Option (A) is correct.
Explanation: Goods used in the production of final goods are known as intermediated goods.

Q.2. Assertion (A): Refrigerator used at home is considered as Capital Good.
Reason (R): Refrigerator is a one-time purchase for the household and will not be used for a very long time. **[OEB]**

Ans. Option (D) is correct.
Explanation: Capital goods includes those fixed or tangible assets that are purchased by producers or businesses to produce goods and services.

Q.3. Assertion (A): Car purchased by the hotel for the purpose of tourism facilitation is a Capital Good.
Reason (R): The car is purchased by the hotel to provide long term service to the consumers. **[OEB]**

Ans. Option (A) is correct.
Explanation: Capital goods includes those fixed or tangible assets that are purchased by producers or businesses to produce goods and services.

Q.4. Assertion (A): Payment of electricity bill by a school is included in the National Income calculation.
Reason (R): Payment of electricity bill is a final good. **[OEB]**

Ans. Option (C) is correct.
Explanation: Payment of electricity bill by a school is included in national income.

Q.5. Assertion (A): Free medical facility by the employer is included in the calculation of National Income.
Reason (R): It forms the part of compensation to employees. **[OEB]**

Ans. Option (A) is correct.
Explanation: Free medical facilities by the employer is included in the estimation of national income because it is a kind of wages in kind and therefore, a part of compensation of employees.

Case Based Questions

[A] Multiple Choice Questions

I. Read the below case and answer the questions that follow:

The country's real gross domestic product (GDP) is likely to expand by 11 percent in the next financial year due to a faster economic recovery and on a low base, says a report. The report by domestic rating agency Brickwork Ratings said economic activities are slowly reaching Pre-COVID levels following the relaxation of the lockdown, except in sectors that remain affected by social distancing norms. "With progress in developing an effective vaccine for COVID-19 and signals of faster-than-expected recovery in the domestic economy, and also supported by a low base, we expect the real GDP to grow at 11 percent in F/Y22, from the estimated contraction of 7 percent to 7.5 percent in F/Y21," the agency said.

According to the first advance estimates of national income released by the National Statistical Office (NSO), the country's GDP is estimated to contract by a record 7.7 percent during the current financial year. **[OEB]**

(a) Real GDP is when the goods and services are produced by all producing units in the domestic territory of a country during an accounting year and valued at _____ (base year's/current year's) prices or constant price.

Ans. Base year's
Explanation: Real GDP does not take into account the effect of price change and hence is measured on base year price.

(b) How is the Real GDP Calculated?

(A) $\frac{\text{GDP}}{\text{GDP Deflator}} \times 100$

(B) $\frac{\text{Nominal GDP}}{\text{GDP}} \times 100$

(C) $\frac{\text{GDP Deflator}}{\text{Nominal GDP}} \times 100$

(D) $\frac{\text{Nominal GDP}}{\text{GDP Deflator}} \times ´100$

Ans. Option (D) is correct.

(c) What will be the growth rate of GDP according to the NSO?

(A) – 7.7% **(B)** +7.7%
(C) +11% **(D)** – 7.5%

Ans. Option (A) is correct.

(d) _____ (Real GDP/Nominal GDP) is the true indicator of the growth of the economy.

Ans. Real GDP
Explanation: Real GDP is the true indicator of the growth of the economy because real GDP does not take into account the effect of price level. It only considers the effect of output only.

[B] Subjective Based Questions

I. While the basic circular flow matrix explains supply and demand in a simplistic economic vacuum, this model doesn't take into account these other key factors of economic systems.

Government sector. Government is an important factor since it both injects money into the flow and also takes money out of it (called "leakage").

Government spending. The government injects money into the economy by buying things from both the product market (like garbage trucks or aircraft carriers), and the resource market (like teachers or fuel). Payments the government makes to both the resource market and the product market are called "government spending." The government uses goods, services, and resources to provide "public goods" like education, roads, and police services. Government spending can also be a public good in itself: examples of this kind of public good include subsidies to businesses (to spur economic development and incentivize manufacturing of a specific type of good) and welfare to households (to help alleviate poverty).

Foreign sector (imports and exports). Instead of money, the foreign sector typically injects goods into the circular flow model in the form of imports and leaks goods in the form of exports.

An economy with the household sector and the business sector on each end and product and resource markets in between is the simplest version of the circular flow model. However, it does not provide a complete picture of the economy. Once the government, financial institutions, and the foreign sector are incorporated into this model, we get a more entire and accurate model of the economic system as a whole. **[OEB]**

(a) How does the household sector and Business sector interact with each other as per the above article?

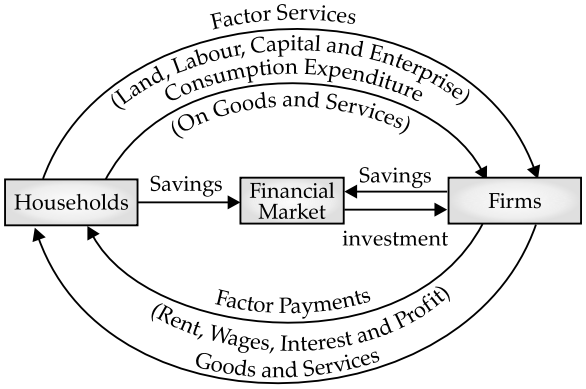
Ans. As per the above article, the household sector provides labour and get factor payment in return of it. The business sector provides goods and services to the households and so get consumption expenditure in return.

(b) How is the government sector important as per the above article?

Ans. As per the article the government sector is important as it injects money into the flow and take out excess money from the flow.

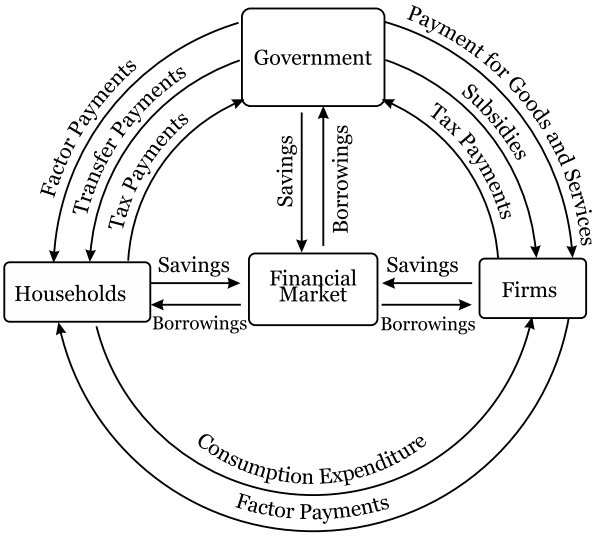
(c) Draw the simplest model of the circular flow of income.

Ans.



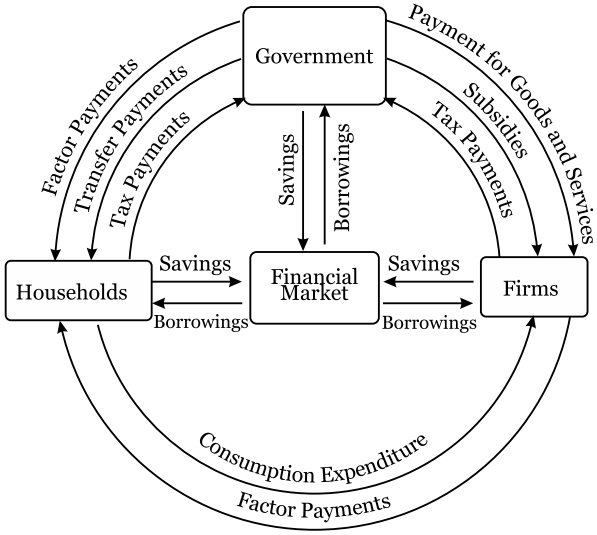
(d) When the government factors as mentioned above are included in the circular flow of Income, what is it called? Draw the circular flow.

Ans. When Government Sector is included it is the Circular flow in a three-sector model.



(e) When the foreign sector is also included what is it called? Draw the model.

Ans. When the foreign sector is also included it is the Circular flow in a four-sector model.



III. The gross domestic product (GDP) of a nation is an estimate of the total value of all the goods and services it produced during a specific period, usually a quarter or a year. Its greatest use is as a point of comparison: Did the nation's economy grow or contract compared to the previous period measured?

Calculating GDP Based on Spending:

One way of arriving at GDP is to count up all of the money spent by the different groups that participate in the economy. These include consumers, businesses, and government. All pay for goods and services that contribute to the GDP total.

In addition, some of the nation's goods and services are exported for sale overseas. And some of the products and services that are consumed are imports from abroad. The GDP calculation accounts for spending on both exports and imports.

Thus, a country's GDP is the total of consumer spending (C) plus business investment (I) and government spending (G), plus net exports, which is total exports minus total imports (X – M).

Calculating GDP Based on Income:

The flip side of spending is income. Thus, an estimate of GDP may reflect the total amount of income paid to everyone in the country.

This calculation includes all of the factors of production that make up an economy. It includes the wages paid to labor, the rent earned by land, the return on capital in the form of interest, and the entrepreneur's profits. All of these make up the national income.

In this income approach, the GDP of a country is calculated as its national income plus its indirect business taxes and depreciation, plus its net foreign factor income.

- How to Calculate the GDP of a Country – Investopedia **[OEB]**

(a) What is the above article talking about? Define the term.

Ans. The above article is talking about GDP.

Gross Domestic Product is the value of all goods and services produced within the territory of the country.

(b) Which two methods have been mentioned in the article? Give the formula for their calculation.

Ans. The two methods that have been mentioned in the above article are:

Expenditure Method: National Income = GDPMP – Depreciation – Net Indirect Taxes + Net Income from Abroad

Income Method: National Income = Compensation to Employees + Operating Surplus + Mixed Income from Self Employment + Net Income from Abroad

(c) Give precautions that needs to be taken care of in the expenditure methods.

Ans. The precautions are:

(i) Expenditure on intermediate goods will not be included in the national income as it is already included in the value of final expenditure. If it is included again, it will lead to double counting of expenditures.

(ii) Transfer payments are not included as such payments are not connected with any productive activity and there is no value addition.

(iii) Purchase of second-hand goods will not be included as such expenditure has already been included when they were originally purchased. Such goods do not affect the current flow of goods and services. However, any commission or brokerage on such goods is included as it is a payment made for productive services.

(iv) Purchase of financial assets (shares, debentures, bonds, etc.,) will not be included as such transactions do not contribute to current flow of goods and services. These financial assets are mere paper claims and involve a change of title only. However, any commission or brokerage on such financial assets is included as it is a productive service.

(v) Expenditure on own account production (like production for self-consumption, imputed value of owner occupied houses, free services from general government and private non-profit making institutions serving households) will be included in the national income since these are productive services. **[Any Two]**

(d) Which method has not been described in the above passage?

Ans. The value – added method has not been described in the above passage.

(e) Write the steps to find the national income by the Value Added Method.

Ans. The steps are:

Step 1: First the producing sectors are classified into primary, secondary and tertiary sectors.

Step 2: The GDP_{MP} is calculated by adding the gross value added at each sector and subtracting the intermediate consumption.

Step 3: The NDP_{FC} is calculated by subtracting the depreciation and Net Indirect tax from the GDP_{MP} calculated in step 2.

Step 4: Finally, NFIA is added to NDP_{FC} to get the NNP_{FC}

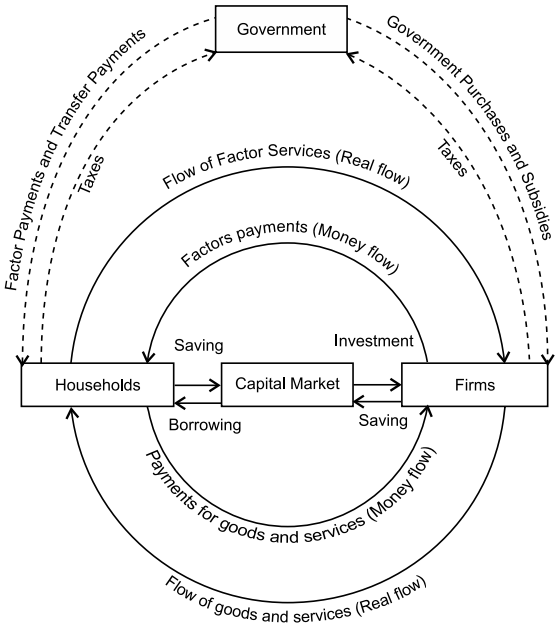
Solutions for Practice Questions (Topic-1)

1 Mark Questions

2. The two sectors of the economy are households and firms.

3 Marks Questions

3.



Circular Flow Diagram : Three Sector Model with Capital Market.

Solutions for Practice Questions (Topic-2)

2 Marks Questions

4. GNP at Factor Cost is the total factor cost of all the goods and services produced in the domestic economy in an accounting year. GNP at Factor Cost is different from National Income. Depreciation is not included while calculating GNP at Factor Cost but it is included while calculating National Income.

3 Marks Questions

5. The difference between intermediate good and final good are:

S. No.	Intermediate Goods	Final Goods
(i)	These goods are used as raw material for the production of other goods during the accounting year.	There goods are not used as raw material for the production of other goods during the accounting year.
(ii)	These goods are used for resale by the firms to make profits during the accounting year.	These goods are not used for resale by the firms to make profits during the accounting year.
(iii)	Value is yet to be added to these goods.	Value is not to be added to these goods.
(iv)	Example: Raw material	Example: Machine purchased for the installation in the factory

Solutions for Practice Questions (Topic-3)

3 Marks Questions

4. (i) $NNP_{FC} = GDP_{MP}$ – Depreciation + Net Factor Income from Abroad – Net Indirect taxes

(ii) Personal Disposable Income = Personal Income – Personal Taxes – Miscellaneous Receipts of the Government Administrative Departments.

REFLECTION

(a) Can you draw and explain the circular flow of income in a two-sector, three-sector and four-sector models?

(b) Can you differentiate between all the basic terms associated with national income?

(c) Can you explain why growth in GDP does not necessarily mean increase in welfare?

(d) Can you calculate National Income using value-added, income and production methods?